



Natural Language Processing: Classification

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Machine Learning and Data-Intensive Systems

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- Pre-deep learning approaches
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- Practical Tips

Extract features, apply model, compare the distribution to the target

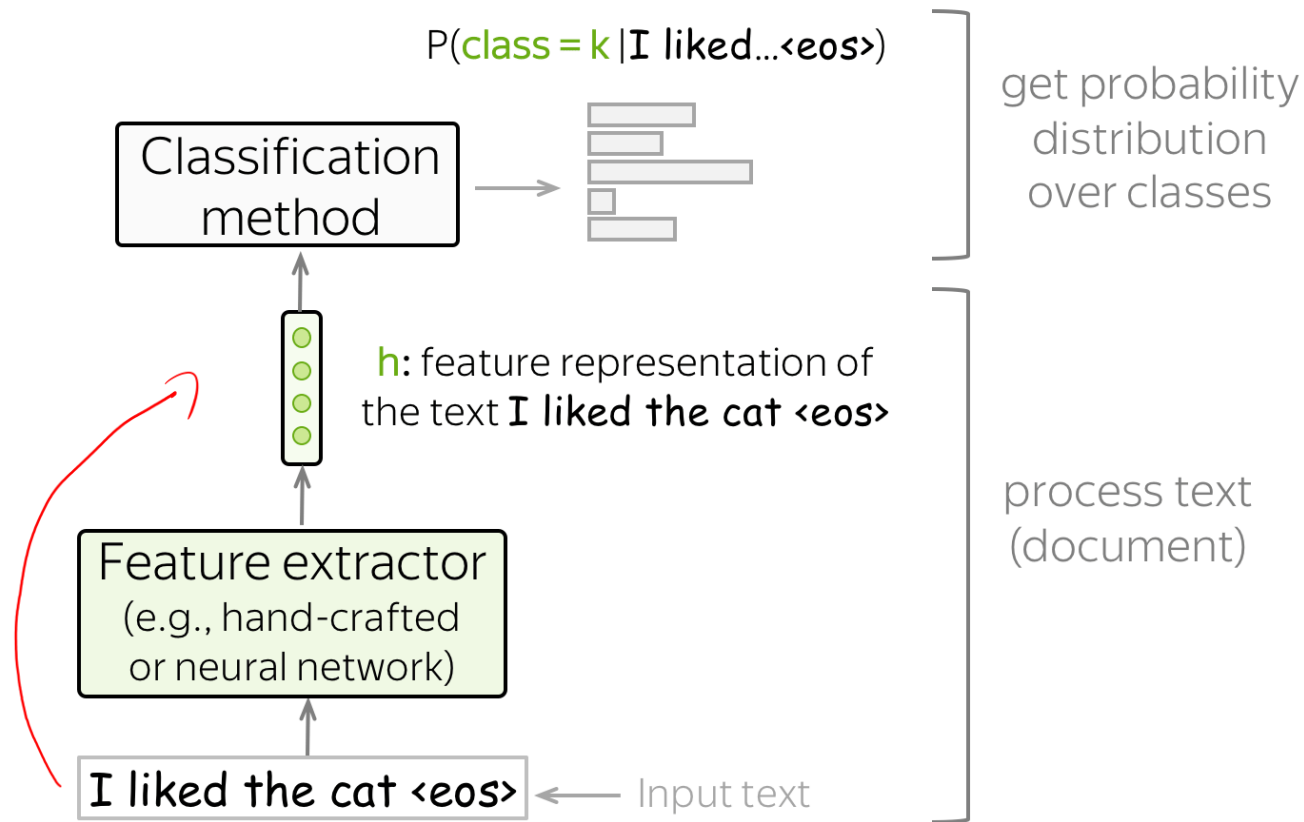
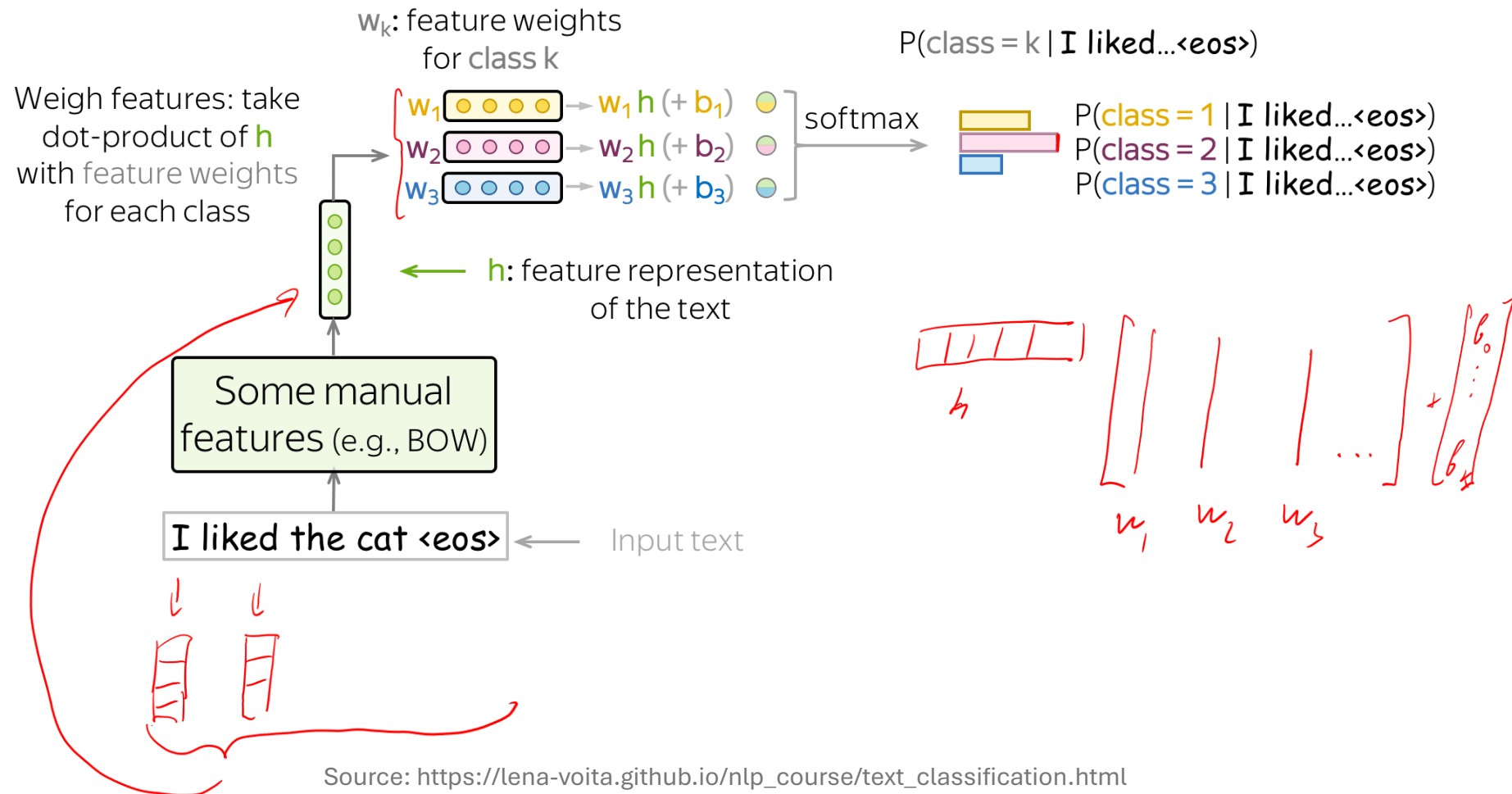




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Train a logistic regression



Loss function: log-loss

Given training examples $x^1, \dots, x^N$ with labels $y^1, \dots, y^N$, $y^i \in \{1, \dots, K\}$, we pick those weights $w^{(k)}$, $k = 1..K$ which maximize the probability of the training data:

$$w^* = \arg \max_w \sum_{i=1}^N \log P(y = y^i | x^i).$$

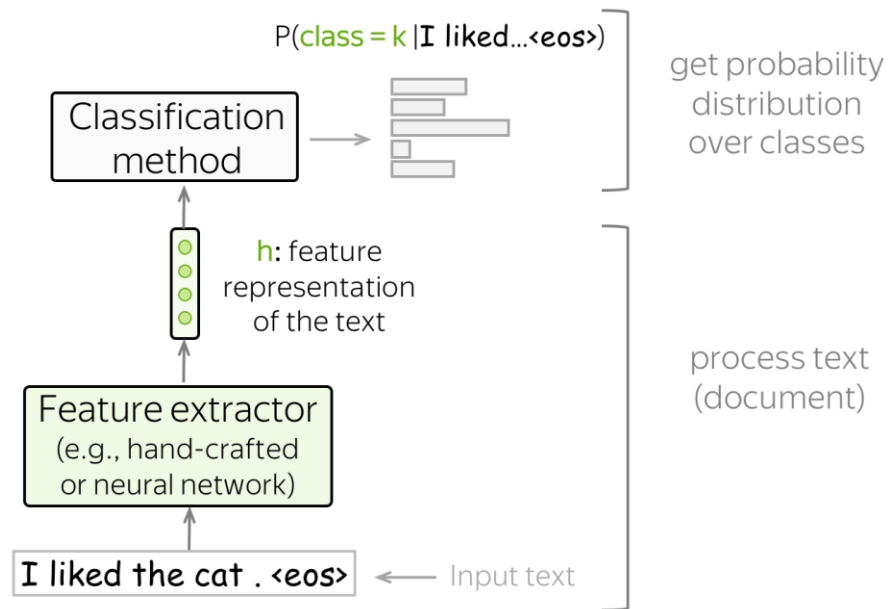


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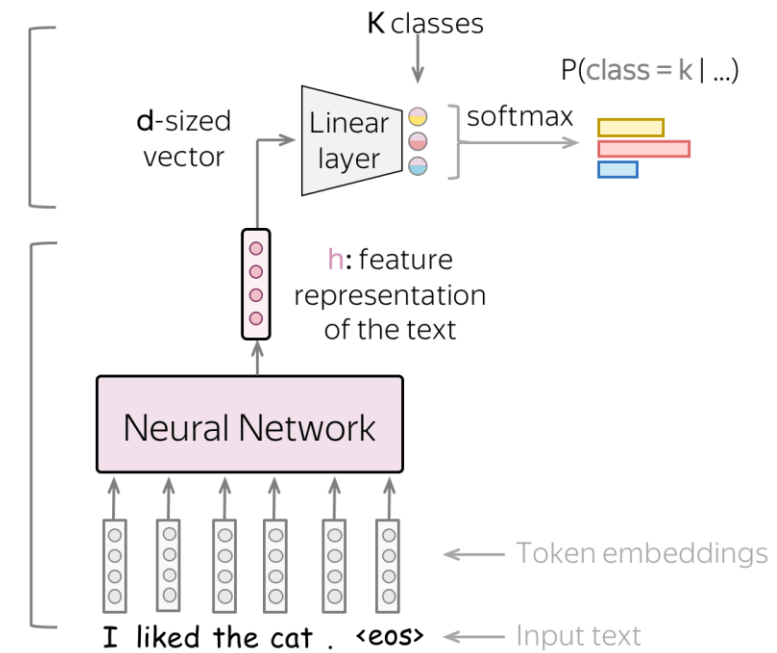
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Overview of DL-based approaches

General Classification Pipeline



Classification with Neural Networks



Overview of DL-based approaches

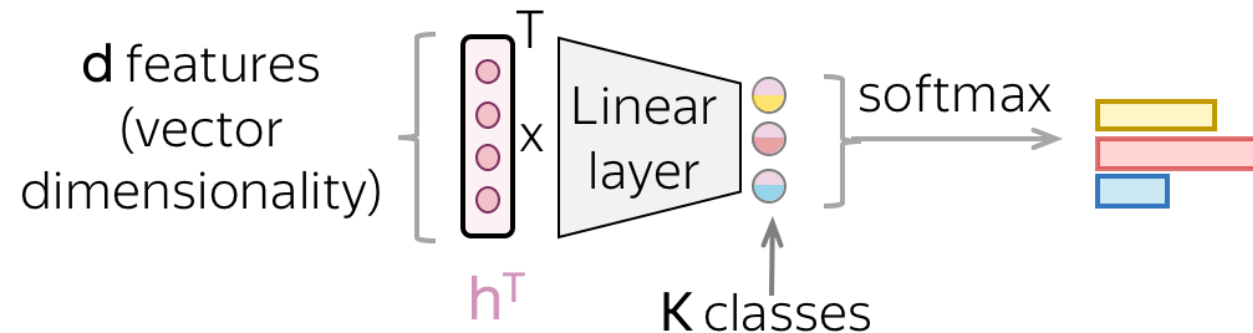
We have:

- h - vector of size d

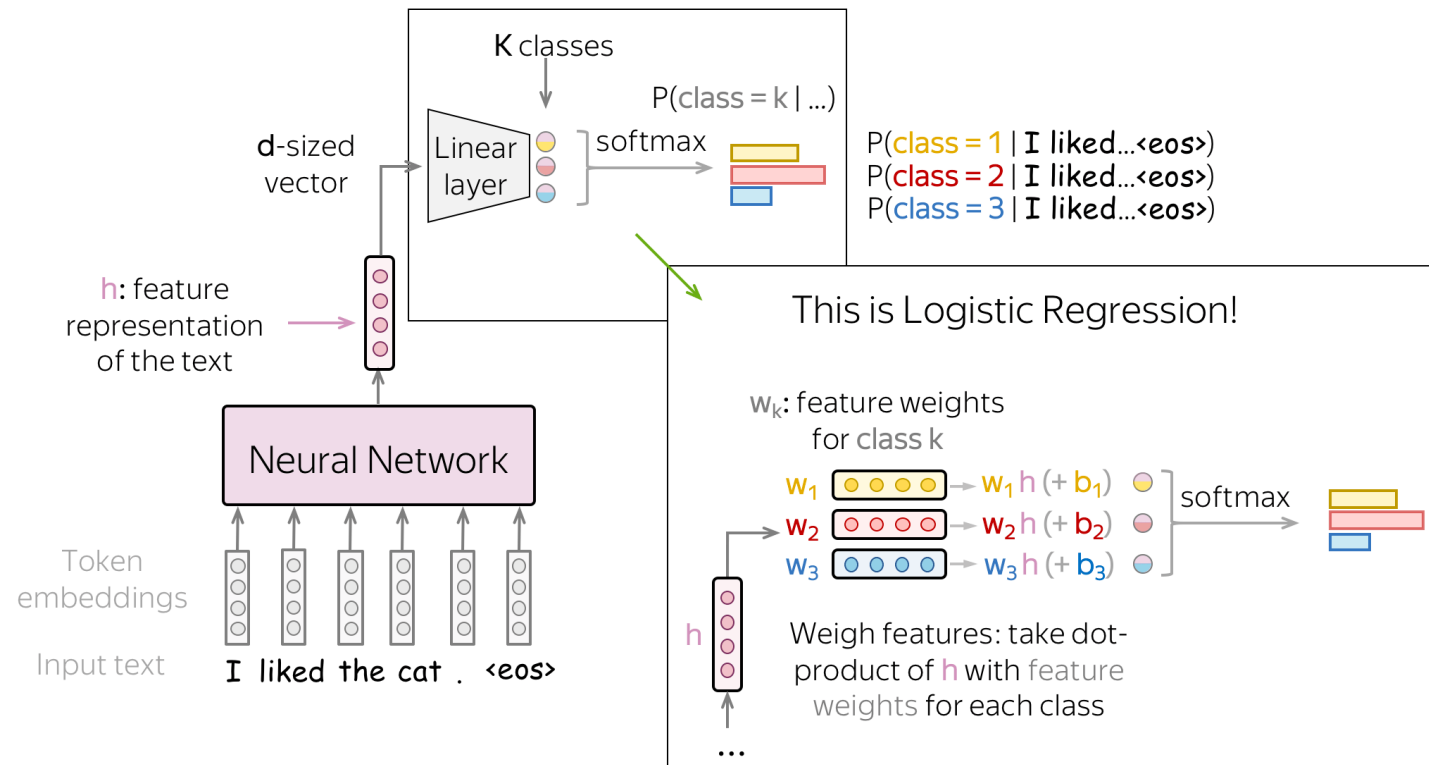
We need:

- vector of size K – probabilities for K classes

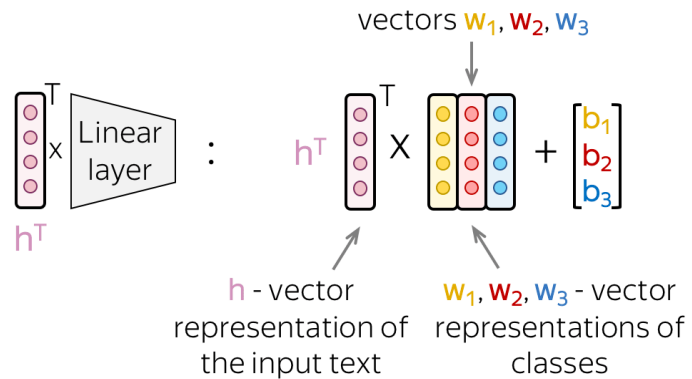
Transform linearly
from size d to size K



Overview of DL-based approaches

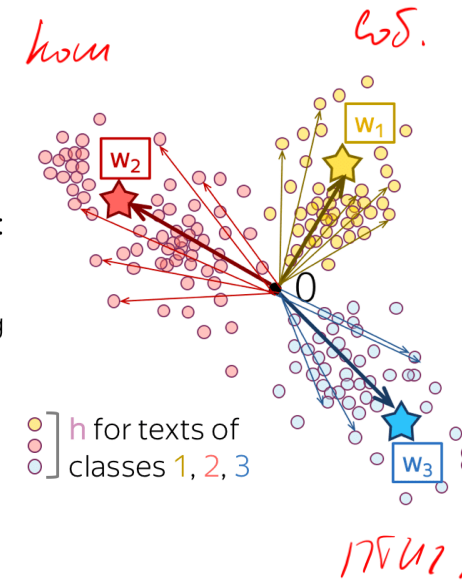


Overview of DL-based approaches



What NN learns (hopefully):

Text vectors point in the direction of the corresponding class vectors



Overview of DL-based approaches

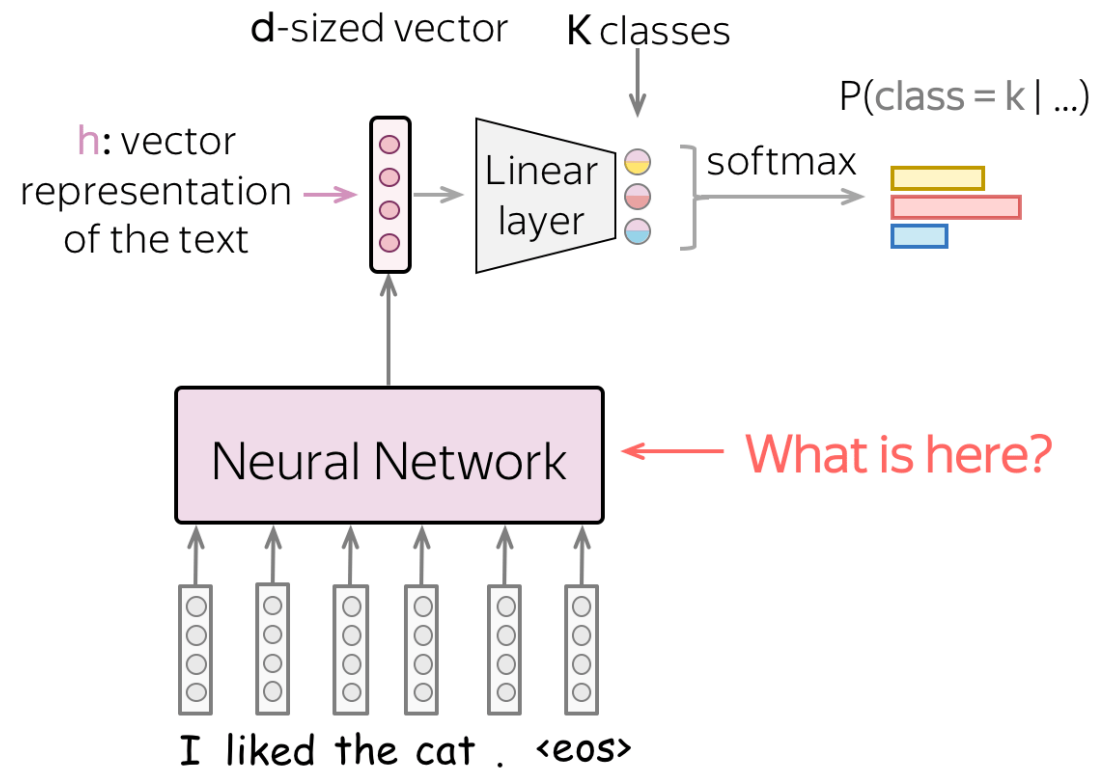




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Vanilla embeddings

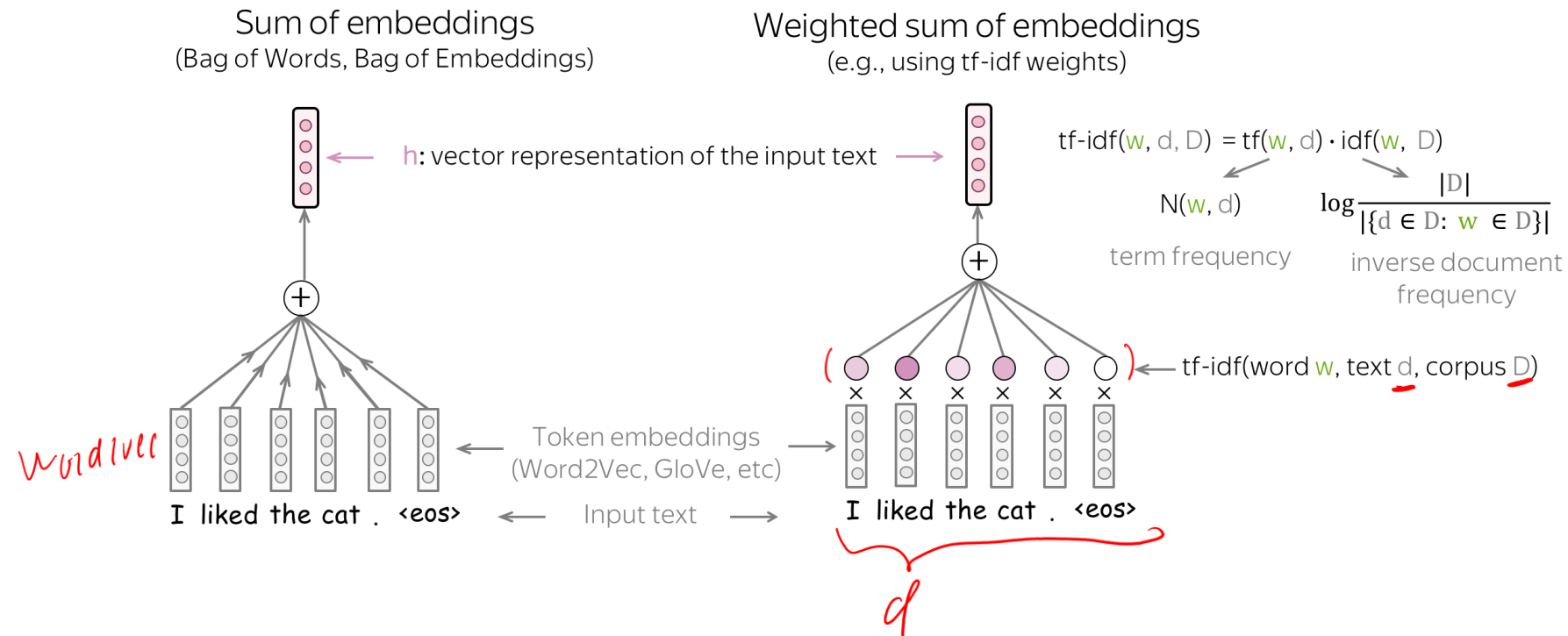
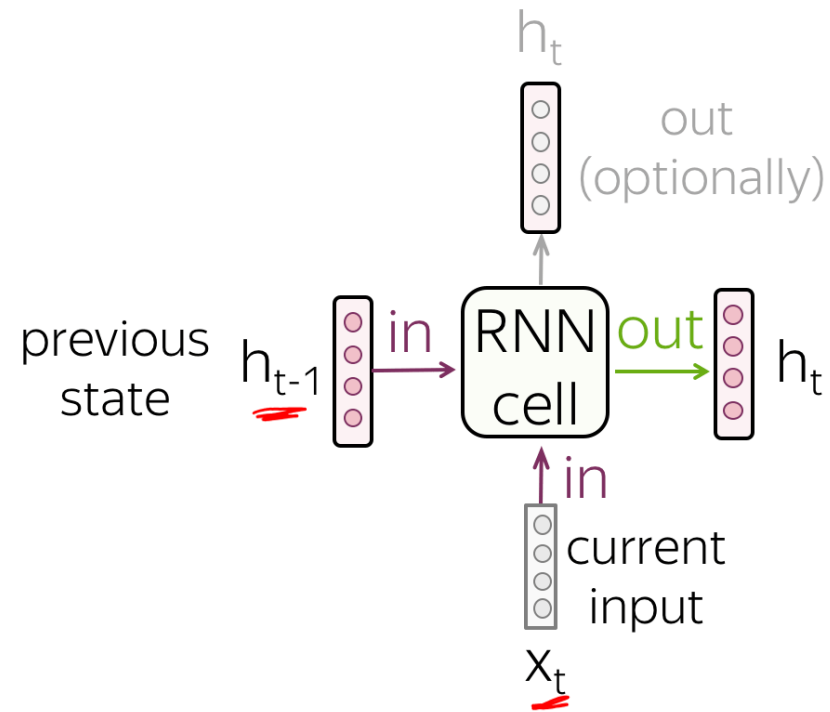




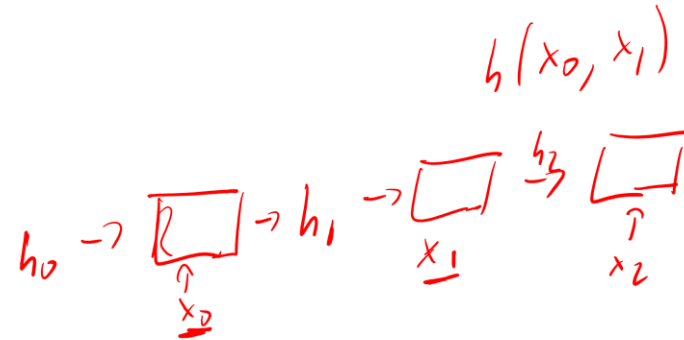
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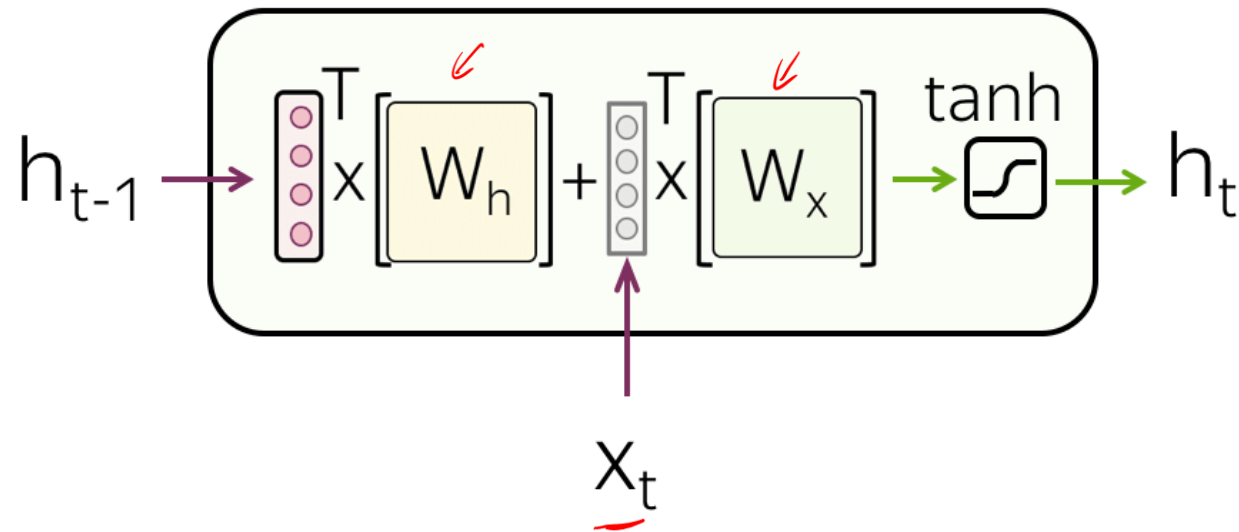
Vanilla embeddings



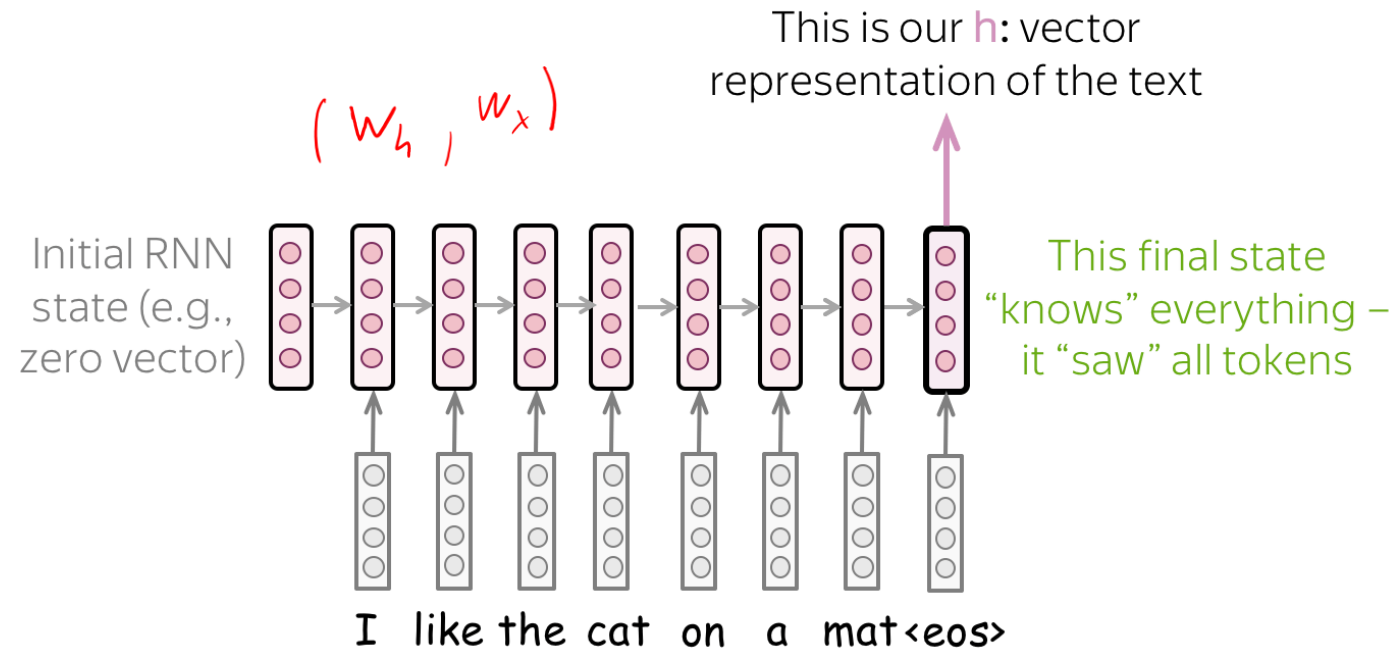
Vanilla RNN



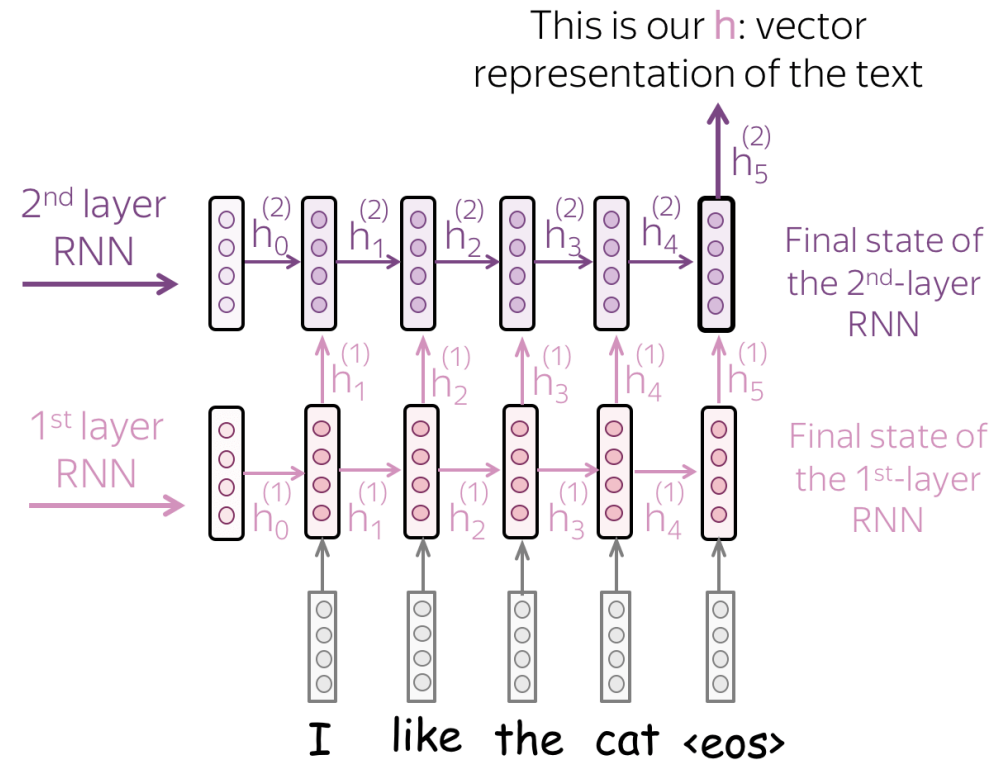
$$h_t = \tanh(h_{t-1}W_h + x_tW_x)$$



Vanilla RNN



Stacked RNN



Bidirectional RNN

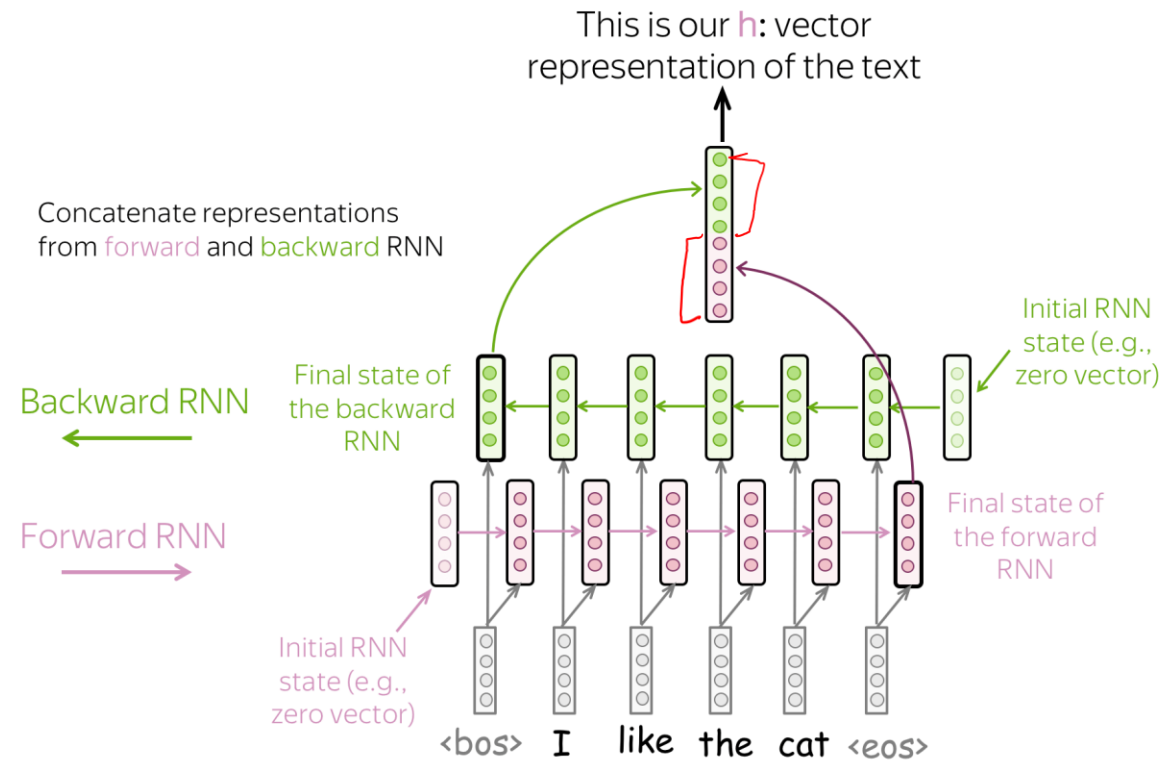




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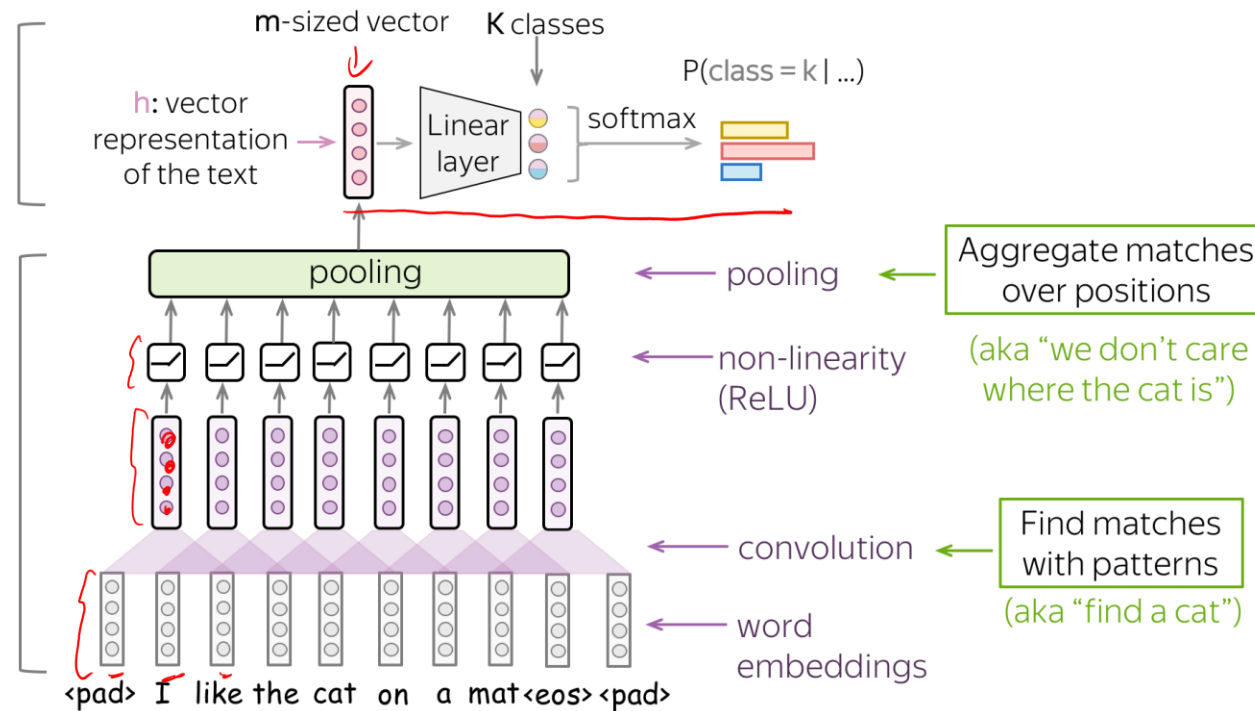
CNN overview

$$\begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix}$$

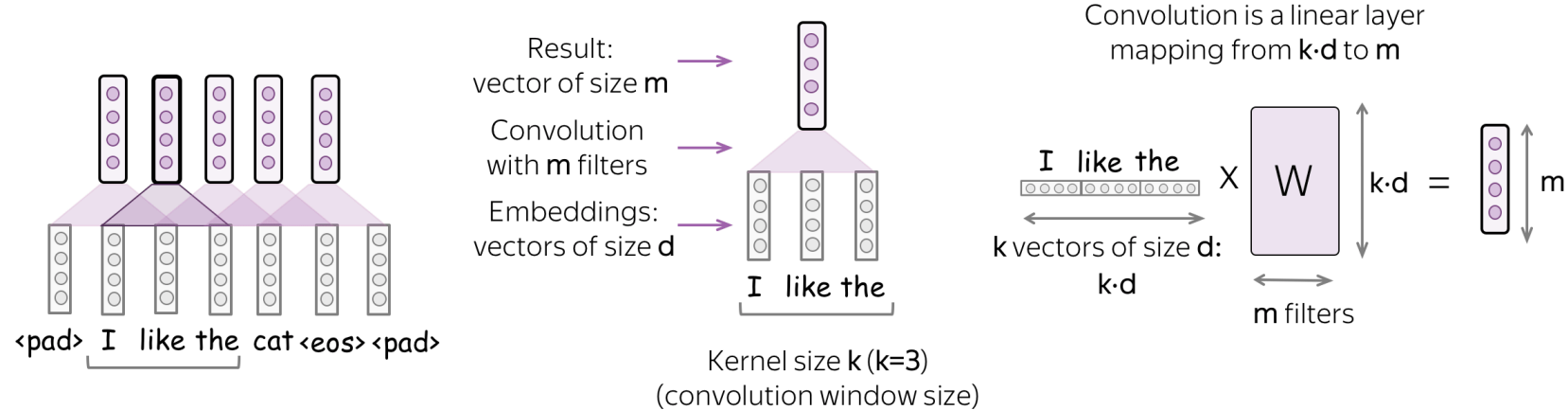


Standard part
(same for all NNs):
get probability
distribution

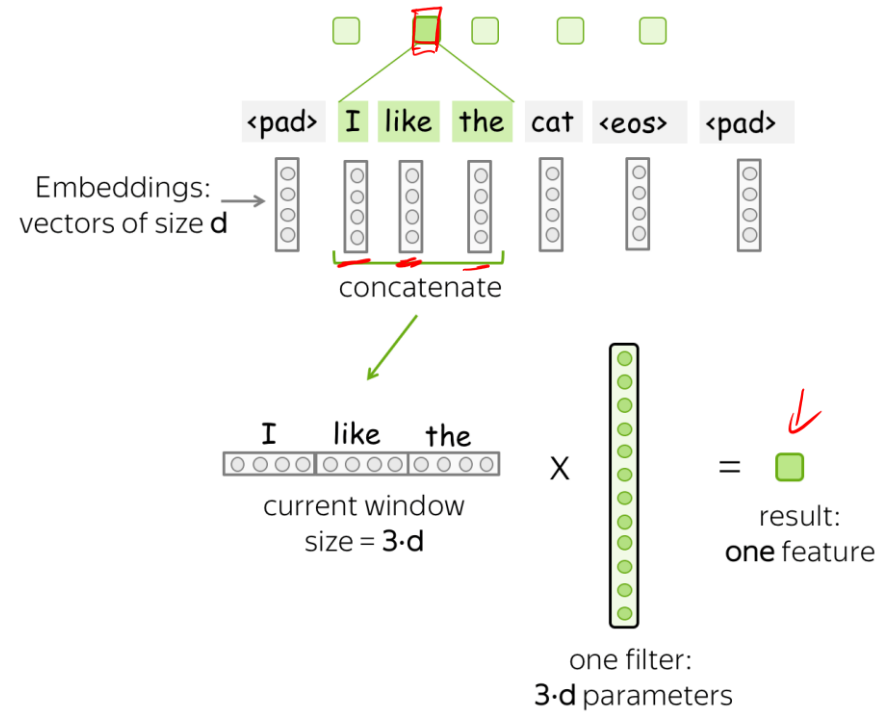
Specific to CNNs:
process text
(document)



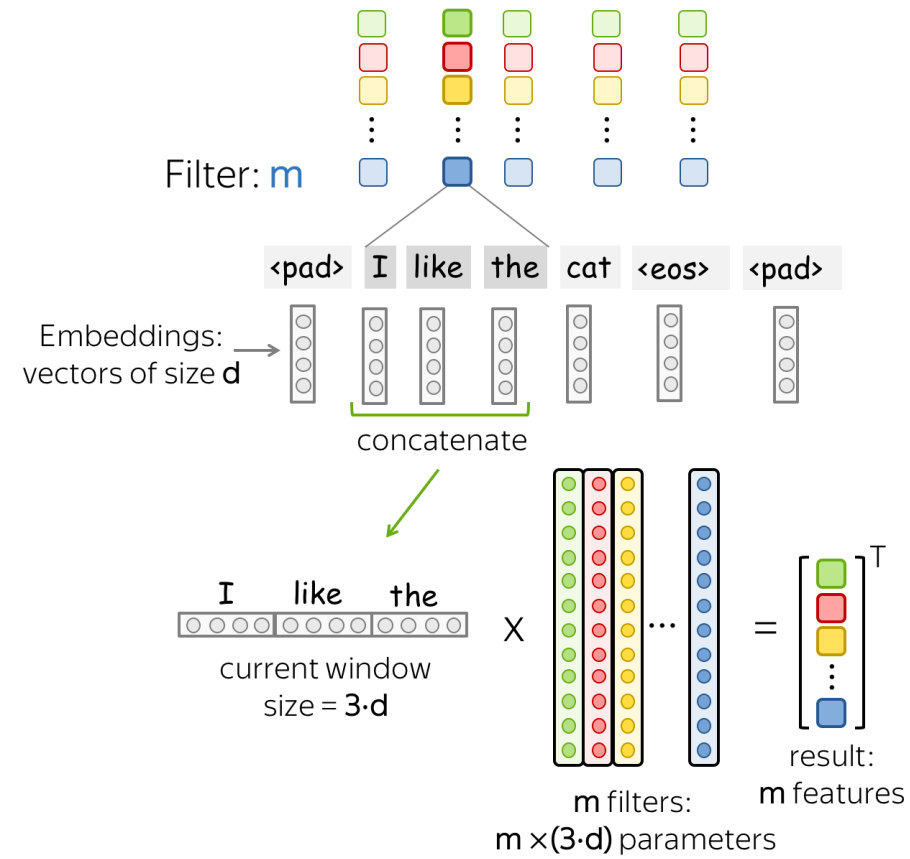
Convolution is a Linear Operation Applied to Each Window



CNN intuition



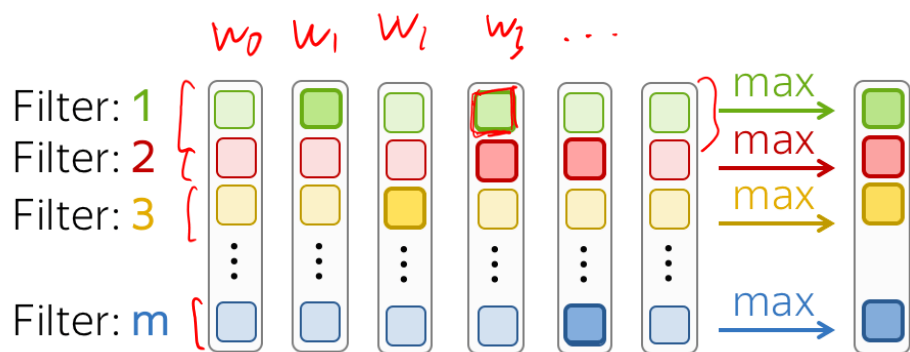
CNN intuition



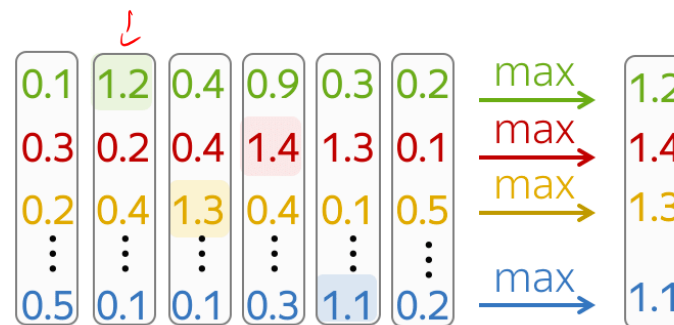
Pooling

$$\text{softmax}([a_0, a_1, a_2]) = v$$

$$\text{softmax}([a_0]) = \frac{\exp(a_0)}{\sum_{i=0}^n \exp(a_i)}$$



Max pooling:
maximum for each dimension (feature)



Larger kernel size for a broader context

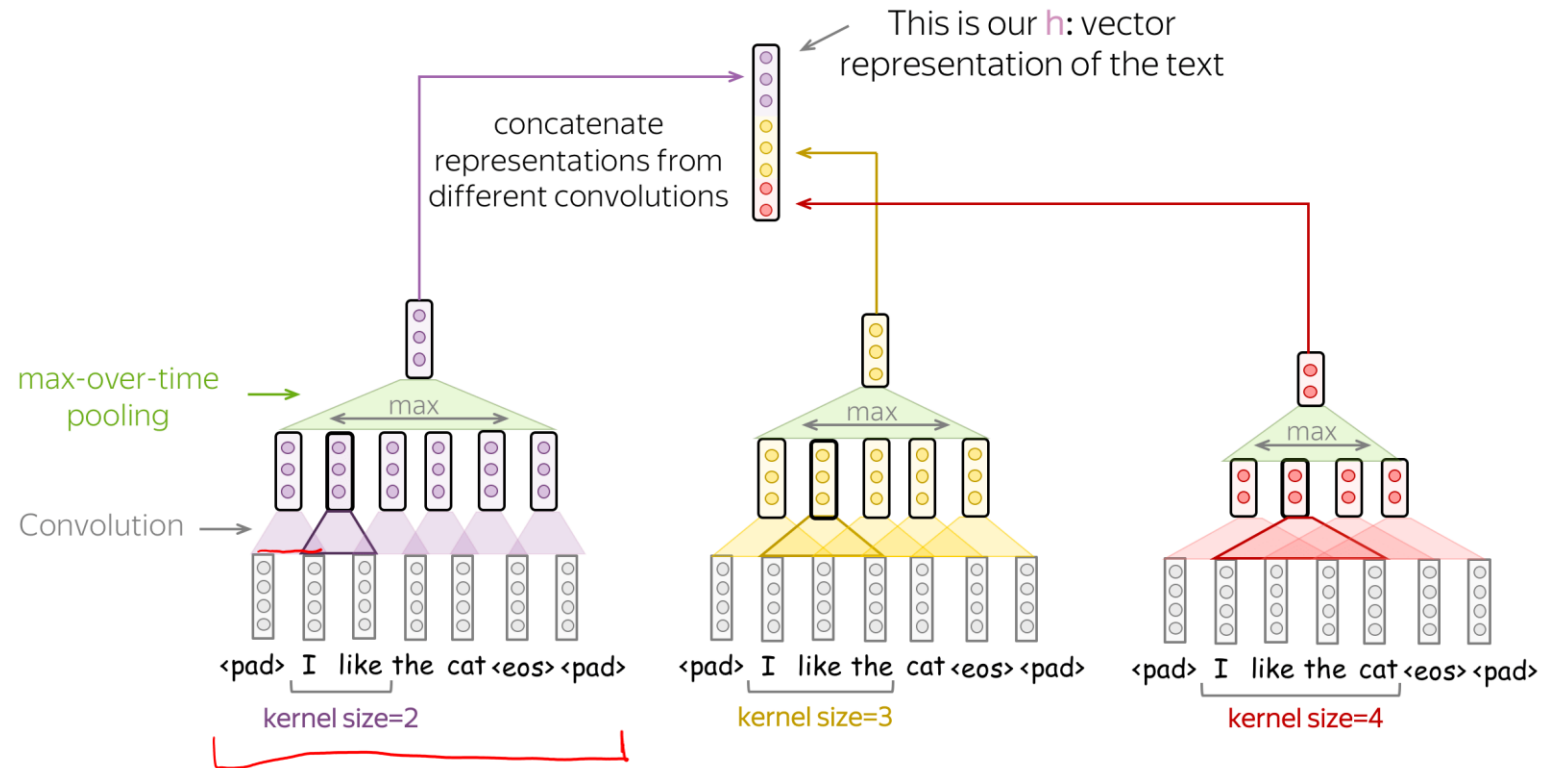
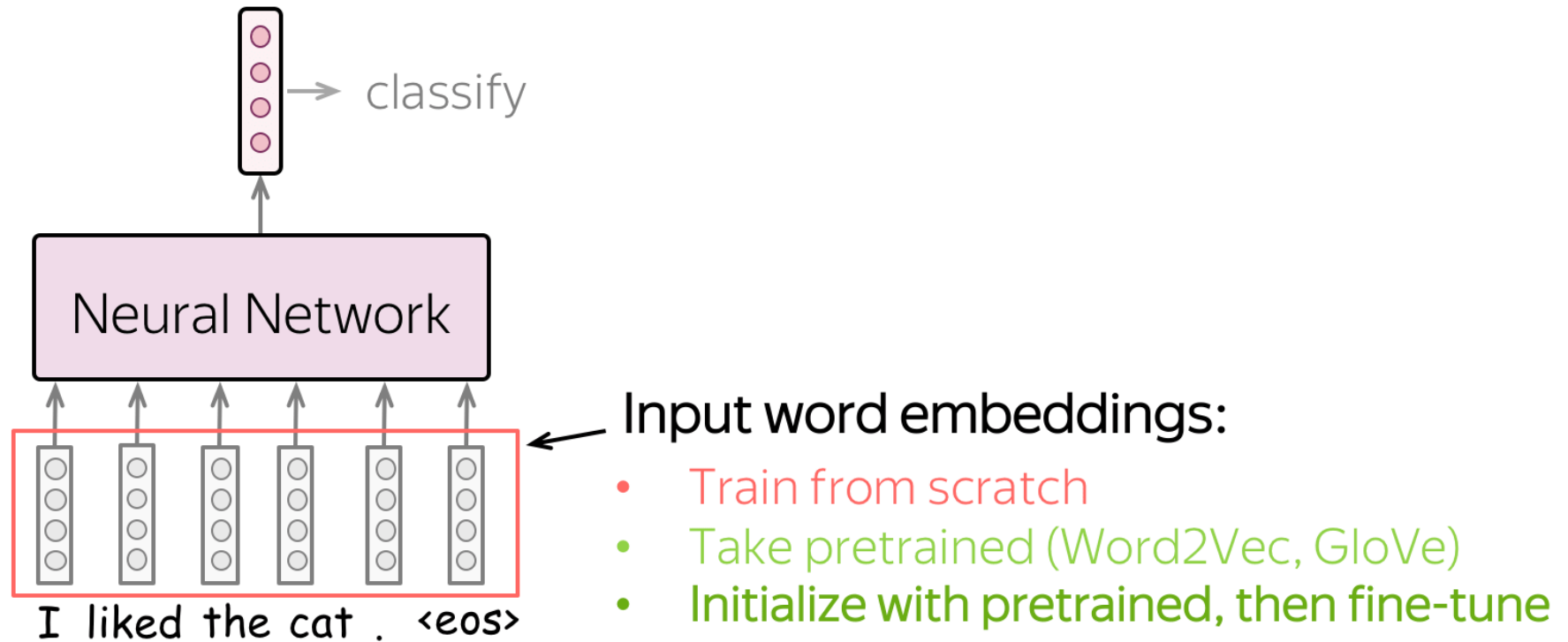




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Larger kernel size for a broader context



Data augmentation

